

Temperature, enzymes and metabolism in *E. coli*

What this model adds to what is published

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The question

- ▶ Metabolism is temperature-dependent; **thermal performance curves are the phenotype**
- ▶ An **etcGEM** predicts a whole TPC from *per-enzyme* thermodynamics — no curve fitting to the shape
- ▶ Three reference points, and every claim here says which one it is relative to:
 - ▶ **Li et al. 2021¹** — the only published etcGEM (yeast)
 - ▶ **Pettersen & Almaas 2023²** — re-ran it and found the inference unstable
 - ▶ **Madkaikar 2023³** — the MRes baseline this group inherited
- ▶ **What is new here: the data that makes the inference identifiable, and the audits that say what a fitted etcGEM does and does not know**

What an etcGEM is

- ▶ A genome-scale metabolic model of ~2 500 reactions, plus three things:
 - ▶ **an enzyme budget** — carrying flux costs protein, and protein is finite
 - ▶ **per-enzyme** $k_{cat}(T)$ — each enzyme's turnover peaks at its own T_{opt} and falls away either side (macromolecular rate theory, curvature $\Delta C_p^\ddagger$)⁴
 - ▶ **per-enzyme unfolding** — the folded fraction collapses at each enzyme's melting temperature T_m
- ▶ Warm the cell and every enzyme moves on its own curve; the LP re-allocates the budget
- ▶ **The organism's TPC is then emergent** — it is an output, not a fitted function

The two temperature responses, per enzyme

Left: turnover $\hat{k}(T)$, each enzyme peaking at its own T_{opt} . **Right:** folded fraction $f_N(T)$, each collapsing at its own T_m .

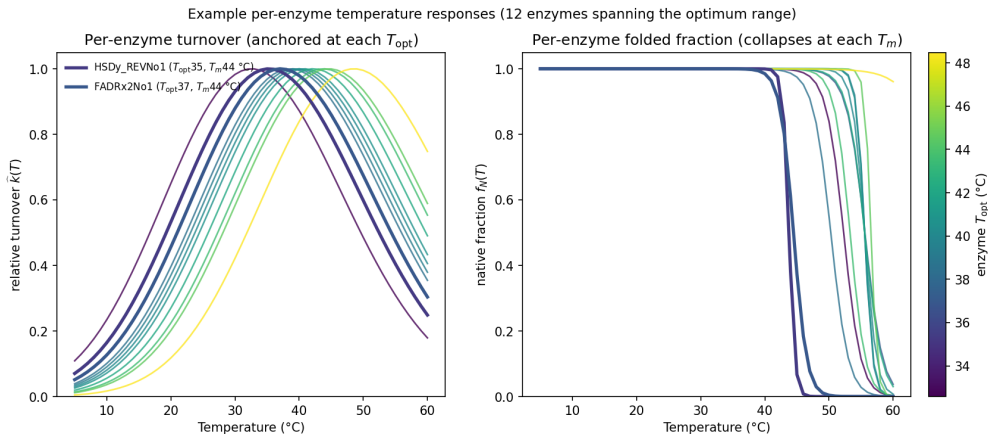


Figure 1: ~12 *E. coli* enzymes. [reports/ecoli_tpc](#)

The state of the art: Li et al. 2021

- ▶ Yeast ecYeast7; **764 enzymes, 2 292 parameters** (T_m , T_{opt} , $\Delta C_p^\ddagger$ each)
- ▶ T_m **measured** (266 of 764); T_{opt} **predicted** from sequence, carried as a wide prior
- ▶ Fitted to **growth and chemostat fluxes**
- ▶ **No gas exchange** — no O_2 , no acetate
- ▶ Headline: ERG1 controls flux at 40 °C, confirmed in the wet lab¹

What one of these curves looks like

Every descriptor a thermal biologist wants is read off **one predicted curve** — and each is a different question of the model.

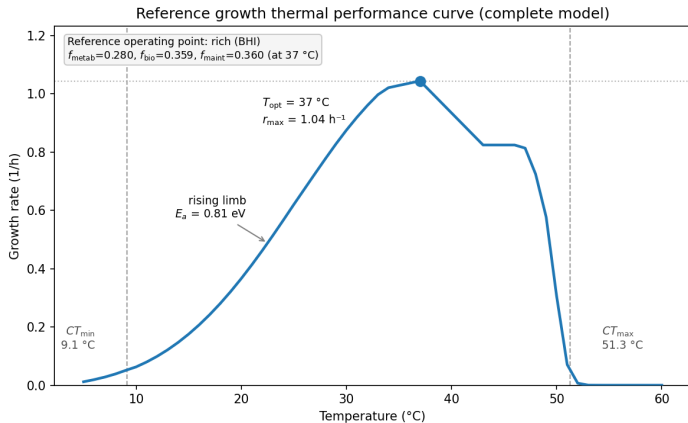


Figure 2: Our *E. coli* model at its reference point. `reports/ecoli_tpc`

The problem the field found: Pettersen & Almaas 2023

- ▶ “the Bayesian calculation method ... is **unstable and unable to estimate the posterior distribution**” — it assumes one peak; the problem has many
- ▶ Of six signature reactions, the one that varies most between **equally-fit** solutions is **ferrocytochrome-c: oxygen oxidoreductase** — “for some particles used extensively, in other cases not at all, still giving rise to approximately the same growth rates”
- ▶ Their diagnosis and their ask:
 - ▶ “the external measurements [are] insufficient to account for the inner workings of the ... cell”
 - ▶ “measurements of proteomics and **fluxomics** will help narrow down the solution space”²

The most under-determined reaction is the O₂-consuming step of respiration, and what they ask for is a measured flux. Hold that thought.

Where we started: the MRes baseline

- ▶ iJ01366, enzyme-constrained with GECKO 3 in MATLAB; **1 366 enzymes**
- ▶ **4 101 thermal parameters** — *more* per-enzyme freedom than Li et al.¹ — with T_m measured for 841 and T_{opt} from a database for 894
- ▶ Same thermal physics as now: two-state unfolding, transition-state k_{cat} with $\Delta C_p^\ddagger$, temperature-dependent maintenance
- ▶ **Not Bayesian**: repeated FBA scored by R^2 against **one** growth curve, the other 25 held out
- ▶ $R^2 = 0.94$; growth TPCs only³

Two things to be clear about: the GEM was later **replaced** (iJ01366 → eciML1515), and the baseline already had per-enzyme parameters — what it had no way to do was **constrain** them.

What the model is now

- ▶ One shared core in Python; **seven organisms** on it (*E. coli* best-constrained)
- ▶ **16 global levers**, deliberately — an identifiable handful in place of 4 101 unconstrainable ones
- ▶ Guarantee against regression: **79/79** structural gates and **60/60** reproduction checks, re-run on every change
- ▶ The figure is **a priori**: nothing was fitted to that curve

Emergent model vs Van Derlinden (K-12 MG1655, BHI; raw absolute rate, nothing fit)

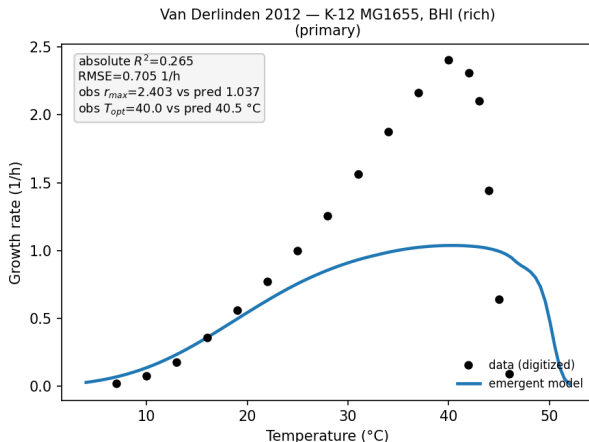


Figure 3: Van Derlinden et al.⁵, K-12 MG1655 in BHI.
reports/ecoli_tpc

The 16 levers — 1: the thermal envelope (6)

Each acts on **every** enzyme at once, on top of the per-enzyme data.

lever	what it does	what it is for
dT_{opt}	shifts every enzyme's T_{opt} by a constant	the database's optima are systematically off
topt_scale	stretches the <i>spread</i> of optima about 30 °C	how synchronised the enzymes are
ΔC_p _scale	scales the curvature of $k_{cat}(T)$	how broad each enzyme's response is
dT_m	shifts every melting temperature	<i>in vitro</i> $T_m \neq$ <i>in vivo</i> inactivation
tm_scale	stretches the <i>spread</i> of melting points	sharp collapse vs gradual shoulder
kcat_scale	scales the <i>level</i> of all turnover	the <i>in vitro</i> $\rightarrow$ <i>in vivo</i> k_{cat} gap ⁶

The 16 levers — 2: budget, maintenance, medium, observation (10)

lever	what it does
σ	average <i>in vivo</i> enzyme saturation — how much of the protein is working
kappa_scale	effective translation efficiency in the biosynthesis cap
f_{metab} , f_{maint}	the proteome split between flux-carrying and housekeeping sectors
ngam_scale, ngam_steepness	size of the maintenance cost, and how fast it rises with T
clearance_mult	how fast the medium is actually drawn down (a <i>medium</i> lever)
resp_scale	mmol O ₂ → measured units: an observation scale, not biology
disc_resp, disc_growth	model discrepancy — how much the model is allowed to be wrong

Which of the sixteen the data can actually move

Three kinds, and the difference matters — not two.

- ▶ **Eight the data move.** dT_{opt} , $topt_scale$, ΔC_p_scale , dT_m , tm_scale , $kcat_scale$, σ , $ngam_scale$
- ▶ **Two are FIXED BY MEASUREMENT** — f_{metab} , f_{maint} . They enter from the measured proteome, matched to medium *and* temperature, and carry **tight priors: measurement wiggle only**. The likelihood not moving them within that width is **the design working**, not a deficiency — growth and respiration are not asked to re-derive a proteomics measurement
- ▶ **Two are genuinely unidentified.** $kappa_scale$ (effective translation demand, derived once at build time — nothing measures it independently) and $ngam_steepness$ (maintenance steepness, *borrowed* from the MRes model). **These** are the ones a measurement would fix
- ▶ **Four are not model levers at all.** $clearance_mult$ is a property of the experiment; $resp_scale$, $disc_resp$, $disc_growth$ are the observation model

Development 1: proteome allocation, from measured proteomes

- ▶ The enzyme budget is split into three **sectors**^{7,8}: **metabolic** (carries flux) · **biosynthesis** (ribosomes) · **maintenance** (chaperones, housekeeping)
- ▶ The split is **not fitted and not hand-set**. It is read from a study that *measured the E. coli proteome at a series of temperatures*, in separate LB / glucose / glycerol series⁹
- ▶ So the model is condition-specific for free: measured ribosome fraction at 37 °C is **0.36 on rich LB** but **0.19 on glucose-minimal** — a rich-medium run automatically gets a higher biosynthesis cap and grows faster
- ▶ The **maintenance** sector carries a measured **2.7-fold heat-stress chaperone rise**
- ▶ New relative to Li et al.¹ and to the MRes³: **neither has an allocation layer at all**

The measured sectors

The chaperone sector (red) ramps sharply above 37 °C — the measured heat-stress response entering as an allocation change, not a fitted parameter.

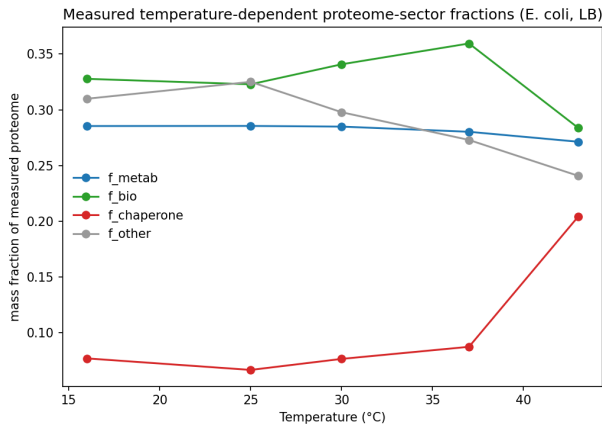


Figure 4: Wang et al.⁹; figure reports/ecoli_tpc

...and this is where we need more data

- ▶ **The measurement is thin where it matters most.** The glucose series was measured at **25, 30 and 37 °C only**; the LB series adds 16 and 43 °C
- ▶ Outside a series' range the allocation is **held at its endpoint** — so above 37 °C on glucose the sector split, and with it the translation cap, **stops varying with temperature**
- ▶ The consequence is visible in the curve: the glucose-minimal prediction is **bit-identical from 37 to 44 °C** (growth exactly 0.524757 h^{-1} at all eight grid points)
- ▶ **That shoulder is the proteomics table running out, not a property of the cell** — and it is why “ T_{opt} ” from such a run is the argmax of a tie

This is the concrete version of “we need more data”: extend the measured glucose series above 37 °C. It is a proteomics experiment, not a modelling problem.

Which lever owns which feature of the curve

- ▶ The heatmap shows **12 rows, not 16**: `clearance_mult` is a property of the experiment, and `resp_scale/disc_resp/disc_growth` are the observation model — none of the four touches the predicted curve
- ▶ Read the rows in four groups:
 - ▶ **catalysis** = $\sigma + \text{kcat_scale}$ — how much working enzyme there is
 - ▶ **kinetic envelope** = $dT_{opt} + \text{topt_scale} + \Delta C_p\text{-scale}$ — where the optima sit and how spread
 - ▶ **stability** = $dT_m + \text{tm_scale}$ — where unfolding takes over
 - ▶ **allocation / maintenance** = $f_{metab}, f_{maint}, \text{kappa_scale}, \text{ngam_}$
- ▶ In lever terms: $dT_{opt}/\text{topt_scale}$ move the **cold** limb and E_a ; $dT_m/\text{tm_scale}$ move the **hot** collapse; $\sigma/\text{kcat_scale}$ move the **height** and nothing else
- ▶ **The result**: r_{max} is owned by catalysis (σ alone has elasticity ≈ 1); T_{opt} and CT_{max} by stability; E_a and niche width by the spread of optima
- ▶ **So a TPC descriptor is not a parameter**. Each one asks the model a different question — which is why fitting to peak height tells you nothing about the hot edge

Elasticity: the numbers

Fractional change in each descriptor per fractional change in each lever, at the operating point. **Deterministic — no intervals.** Bold is $|E| > 0.3$.

lever	T_{opt}	r_{max}	CT_{max}	niche width	E_a
dT_{opt}	0.00	-0.09	0.00	-0.34	0.26
topt_scale	0.00	-0.23	0.00	-1.05	1.41
ΔC_p _scale	0.00	-0.02	0.00	0.29	0.22
dT_m	0.13	0.09	0.13	0.18	-0.24
tm_scale	-0.26	-0.14	-0.16	-0.21	0.24
kcat_scale	0.00	0.40	0.00	0.07	-0.37
σ	0.00	1.00	0.00	-0.11	0.36
f_{maint}	0.00	-0.56	0.00	0.06	-0.37
ngam_scale	0.00	-0.00	0.00	-0.01	0.01
the three zero rows	0.00	0.00	0.00	0.00	0.00

...and the same numbers as a pattern

The three zero rows are zero for **different reasons**: kappa_scale and ngam_steepness unidentified, f_{metab} fixed by measurement.

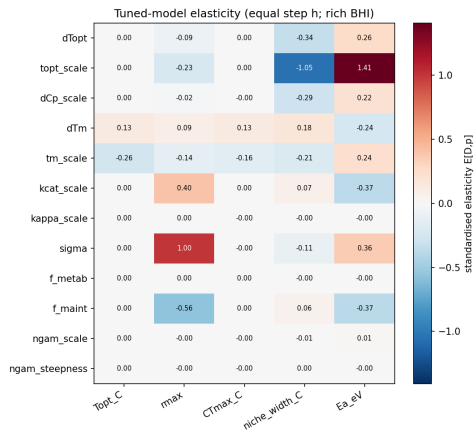


Figure 5: Elasticity heatmap, $h = 0.10$. reports/ecoli_tpc

Development 2: overflow metabolism — what it is

- ▶ Grow *E. coli* fast enough and it **excretes acetate while oxygen is still available** — it ferments and respire at once, wasting carbon. The bacterial analogue of the Crabtree effect
- ▶ Measured law: excretion is zero below a growth-rate threshold, then linear.
Threshold $\approx 0.76 \text{ h}^{-1}$, slope $\approx 21 \text{ mmol gDW}^{-1} \text{ h}^{-1}$
- ▶ **Why any of this matters for a temperature model:** an enzyme-constrained TPC fixes the growth rate but *not* the internal flux distribution, so **O₂ and CO₂ exchange are under-determined** — the model can hit the right growth rate by respiring or by fermenting
- ▶ That is the *same* degeneracy Pettersen & Almaas found by FVA on the oxidase reaction², seen from the other side
- ▶ **Until it is pinned, an etcGEM cannot predict a gas flux at all**

...and how it is made to emerge

Four ways to pin it — Parsa Amirmoeini's configurations A–F, ported here and gated.

	mechanism	imposed or emergent
A	cost the membrane carriers: transporters get the same thermal envelope as every other enzyme	emergent ceiling
B	a total-carbon cap ,	emergent
	$\sum_i n_{C,i} v_i \leq c_{max}$	
C	the measured acetate line written in as a constraint ⁷	imposed
D	the carbon cap, oxidase split left free — the LP picks <i>bd</i> vs <i>bo3</i> per temperature	emergent, T-dependent
E/F	an ETC membrane-area	emergent

Why configuration D is the interesting one

- ▶ **Nothing is imposed on the acetate branch.** Overflow falls out of the protein budget plus a carbon budget — the model is not told that acetate appears above 0.76 h^{-1} , it produces it
- ▶ The **terminal-oxidase choice becomes a prediction:** *bd* and *bo3* have different melting temperatures, so which one the cell uses shifts with temperature on its own
- ▶ c_{max} is **not our number:** Parsa's own sweep (60 / 100 / 120 / 180 mmol C gDW⁻¹ h⁻¹) concludes 100–120 is the sweet spot — *“a tight cap gives respiratory gas but a flat growth plateau; a loose cap gives a peaked TPC but fermentative gas”*. 120 adopted; 450 on LB
- ▶ Data behind the layer: the **acetate law**⁷, the **membrane geometry** (Szenk et al., surface-to-volume $4.5 \mu\text{m}^{-1}$), and Parsa's own **respirometry** for the fits
- ▶ **The advance over Li et al.¹ and the MRes³: both stop at growth.** This turns O_2 , CO_2 and acetate into predictions that a measurement can check — and Parsa's respirometry is that measurement

Configuration D: the growth curve it predicts

The medium ladder is the measured sector split at work: r_{max} **1.19** (glucose) to **2.02** **h⁻¹** (BHI/LB), T_{opt} 38–40 °C. **Model output; no measurement plotted.**

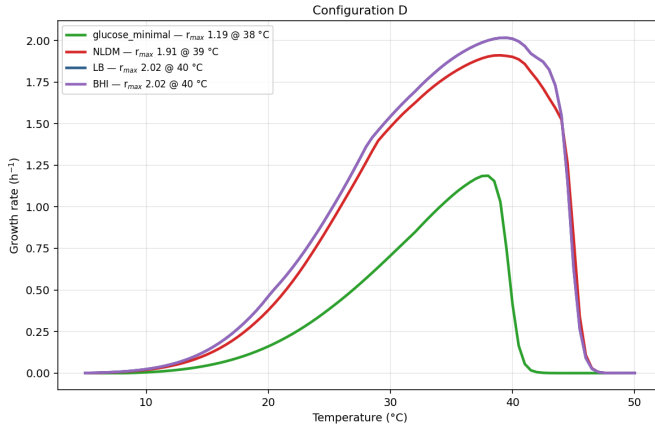


Figure 6: reports/ecoli_gasflux

...and the gas exchange it predicts

Right-hand panel: $RQ \approx 1$ around the optimum — respiratory, not fermentative.
Model output; no measurement plotted; mechanism, not interval.

Configuration D — gas exchange (collapsed tail masked)

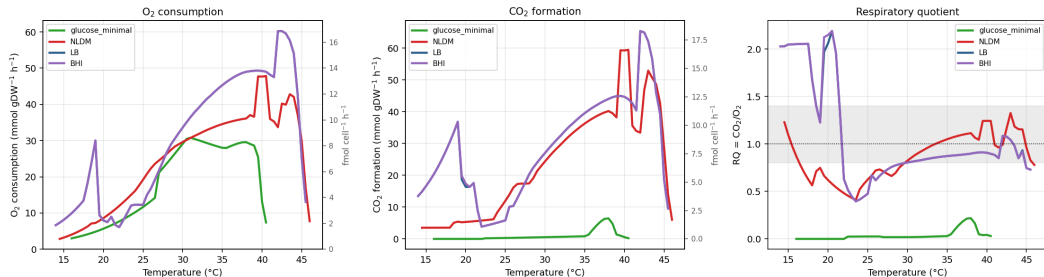


Figure 7: reports/ecoli_gasflux

The data nobody else has

- ▶ Parsa's respirometry: **O₂ flux and growth per cell, 15–50 °C, three media** (R2A, LB, M9), ~60 replicate runs each
- ▶ This is exactly the **fluxomics on the O₂-consuming reaction** that Pettersen & Almaas² name as the missing data
- ▶ And it is a **measurement**, not a model output — which is what makes it able to break the degeneracy
- ▶ Ported and **gated**: Parsa's own numbers reproduce here (60/60 on the configurations, 10/10 on the respirometry fits)
- ▶ Look at the shape: **respiration peaks 3–5 °C above growth** (43–45 °C vs ~40 °C) and is still climbing where growth has begun to fail

Measured growth and respiration, three media

Respiration peaks 3–5 °C above growth and is still climbing where growth has begun to fail. Faint points are replicate runs; bars s.e.m.

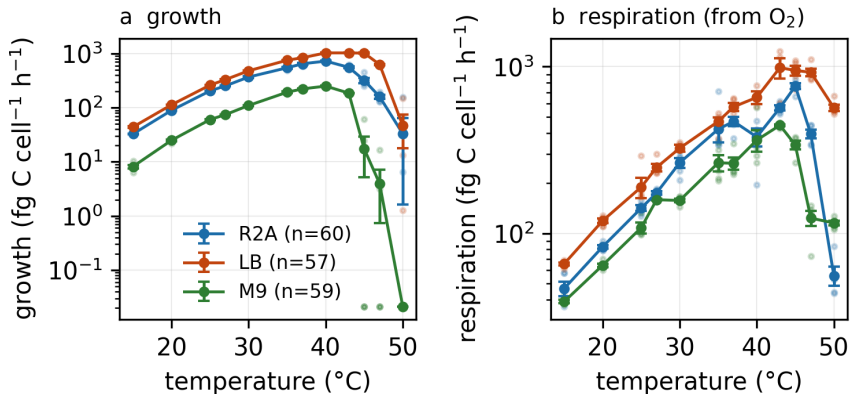


Figure 8: strains/eciML1515/respirometry/

...and what it costs the cell

- ▶ **Carbon-use efficiency** — the share of assimilated carbon kept as biomass rather than respired
- ▶ Flat at **~0.6** (R2A, LB) and **~0.4** (M9) from 25 to 40 °C
- ▶ Then a **cliff**: 0.29 / 0.52 / 0.05 at 45 °C
- ▶ Medium sets the *level*; temperature sets the *cliff*
- ▶ CUE is listed in our own *E. coli* paper as **future work**. It is measured, in three media, and not yet folded in

Measured carbon-use efficiency

Flat from 25 to 40 °C, then a cliff — and the medium sets the level, not the shape.

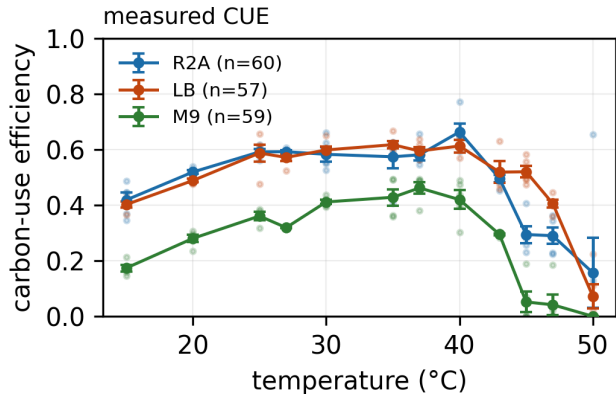


Figure 9: strains/eciML1515/respirometry/

The statistical problem, in plain terms

What everyone is trying to do: the per-enzyme parameters are uncertain, so treat them as unknowns and ask the data which values are consistent with it — a *distribution* over parameters, not one best guess.

- ▶ **Li et al.** used **SMC-ABC**: draw parameter sets from the prior, simulate, keep the ones whose predictions land close to the data, narrow the tolerance, repeat. Reported a posterior over 2 292 parameters¹
- ▶ **Pettersen & Almaas** re-ran it **from different random seeds and got different answers**. The method assumes the good parameter sets form a single hill; in this problem there are many. They swapped in an **evolutionary algorithm** to collect a *diversity* of good solutions instead — then showed those solutions disagree wildly on internal fluxes²
- ▶ **What we do differently: 16 parameters, not 2 292** — small enough for a proper MCMC sampler (an ensemble sampler) rather than an approximate one — and a likelihood that includes the **measured respiration**, not growth alone

...and the three things that went wrong

1. **The chains are too short.** Every chain in this family — ours and Parsa's — ran about **9 autocorrelation times** where the standard is ≥ 40 . So **no interval on any parameter is available**, and none appears in this deck. Fixing it is arithmetic: $4\text{--}5\times$ the chain length
 2. **The surface itself is not smooth.** We stopped blaming the sampler and looked at the likelihood: 22 line scans, 902 model builds, **12 of 22 lines jump by $>20\%$ of their range on a small step**. Not the solver — unchanged at 10^{-9} tolerances and on a different algorithm
 3. **The cause is the respiration term, and it is the degeneracy again.** At cold temperatures the O_2 flux at the growth optimum **jumps between equally-optimal solutions** — so the prediction moves discontinuously while the growth rate does not. *Exactly* what Pettersen & Almaas saw by FVA², here as cliffs in a likelihood
- **Three fixes, each for a named mechanism:** a **tie-break** to choose one point on the degenerate face; a **variance floor** so the likelihood is not shocked by a jump the model cannot avoid; a **continuous support** clamp at the growth threshold. Cliffs cut $\sim 10\times$. **Still not smooth, and we say so**

The surface, scanned

One panel per parameter line — read the pattern, not the axes: parabolas broken by cliffs, and two panels flat all the way across.

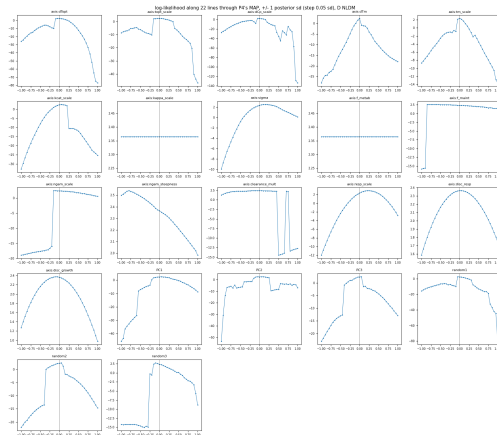


Figure 10: 22 line scans, 902 model builds. reports/P9_surface

How many answers are there? — and two retractions

- ▶ We then mapped the **basins**: run the optimiser from many starting points, cluster where it lands, and ask how many distinct good solutions there are
- ▶ **Result: one live basin and one dead one**, of twelve converged endpoints. **Not** the many-modes picture Pettersen & Almaas² found in yeast — but also **not yet a posterior**
- ▶ **Retraction 1 (ours)**. We had concluded the model *cannot* fit these data while honouring the measured melting temperatures. **It can**: with both melting-temperature levers pinned to the measurement the fit converges and the organism grows at 1.66 h^{-1} — 7.6 log-likelihood units worse, and pressed against two prior bounds
- ▶ **Retraction 2 (ours)**. The test we used to claim the optimiser reaches the top of the basin was **partly an artefact**: five of twelve endpoints park a temperature on an internal threshold that acts as an attractor

Where the posterior stands

- ▶ **Two runs are in flight.** The stopping rule was fixed **before** they started: chain length > 40 autocorrelation times **and** effective sample size ≥ 200
- ▶ **Nothing is quoted from them here** — that is the whole point of pre-registering the criterion
- ▶ What we will be able to say when they land: an interval on each of the sixteen levers, conditional on the mode, and a posterior-predictive band on the TPC and on the gas fluxes
- ▶ What we will **still** not know: `kappa_scale` and `ngam_steepness` — those need a measurement, not a longer chain

The honest position today: mechanisms that reproduce, a likelihood we understand, and no intervals.

Auditing the published model

- ▶ We built an audit for a defect class we had found in our **own** models — a free ion path that lets ATP synthase run without the respiratory chain paying for it — and ran it on Li et al.'s deposited model¹
- ▶ **The defect is absent.** Their respiratory chain supplies **93 %** of ATP synthase's protons; all twelve uncostered proton-movers run the dissipative way; **no free path exists at any length**
- ▶ Then we repeated one of our own structural findings at **their published posterior** — 100 models, their code, their parameters: changing which constraint binds widens the top of the TPC from **1.5 to 7.3 °C in 93 %** of their models
- ▶ **But CT_{max} also moves (4.6 °C).** In *our* model it did not. **The asymmetry is not the clean thing we thought it was**
- ▶ Register: an audit applied to the reference implementation — not a critique of a paper

The asymmetry at Li et al.'s own posterior

Left: the top of the curve widens from ~ 1.5 to ~ 7.3 °C. **Right:** T_{opt} moves further than CT_{max} in 92 % of their models — but CT_{max} is **not** immune.

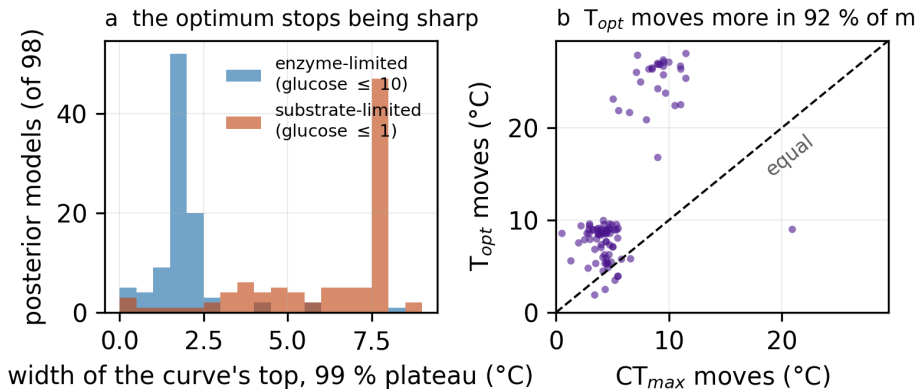


Figure 11: reports/Y2_regime_posterior

The awkward one: our model needs a -4 K stability shift

- ▶ Every fit of the *E. coli* model wants melting temperatures **3–4.5 K below a measured meltome**¹⁰
- ▶ Two readings, and we screened them: **biology** (*in vitro* T_m is not *in vivo* inactivation) or **our parameterisation** (16 global levers where Li et al.¹ had 2 292)
- ▶ **The shift is not uniquely required** — an equally good, living fit exists at zero shift, for 0.08 log-likelihood units
- ▶ **But catalysis is not what pays for it.** The compensator is tm_scale , the *spread* of melting temperatures. Both solutions put the least-stable few per cent of enzymes **4–6 °C below measurement**
- ▶ So the **number** is a parameterisation artefact; the **contradiction with the meltome's low tail** is not. Per-enzyme $k_{cat}(T)$ would not fix it — and the predictor we would have used has no usable optimum for 97 % of these enzymes

Settled · in flight · needs measurement

- ▶ **Settled.** The shared core and the seven-organism port, gated. The mechanisms: measured proteome sectors, overflow emerging from a carbon cap, the ETC area budget. The audits — including the reference implementation passing ours
- ▶ **Settled, and uncomfortable.** No interval from the older *E. coli* fits is available. The likelihood surface is structurally rough. The meltome's low tail is contradicted by every fit
- ▶ **In flight.** The posterior, on a pre-registered convergence criterion
- ▶ **Needs measurement — not more modelling.** Proteome allocation **above 37 °C on glucose** — the measurement exists, its *range* does not, and that is where CT_{max} is set; maintenance vs temperature from a low-dilution chemostat (ngam_* are borrowed); *in vivo* stability for the enzymes that set the thermal limit
- ▶ **Never yet tested.** Nothing in this framework has been asked to predict data it was not fitted to. Cooper et al.'s evolved *E. coli* lines¹¹ are the obvious holdout

Three questions for the group

1. **Is a few-kelvin gap between an *in vitro* meltome and *in vivo* functional inactivation expected?** If it is, the model is asking the meltome the wrong question and we should say so. If it is not, the model is wrong somewhere we have not looked
2. **Respiration peaks 3–5 °C above growth, and carbon-use efficiency falls off a cliff at 43–45 °C.** Is that a general property of bacterial thermal performance, or a feature of these three media?
3. **What would you accept as a real test?** Our candidate is predicting how a TPC *shifts* after 20 000 generations of thermal selection¹¹. Is there a better one?

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